

Лабораторная работа 8

Адресация IPv4 и IPv6. Настройка маршрутизации

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1. Информация

Раздел 1

1. Информация

1.1 Докладчик

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- д.ф.-м.н., студент



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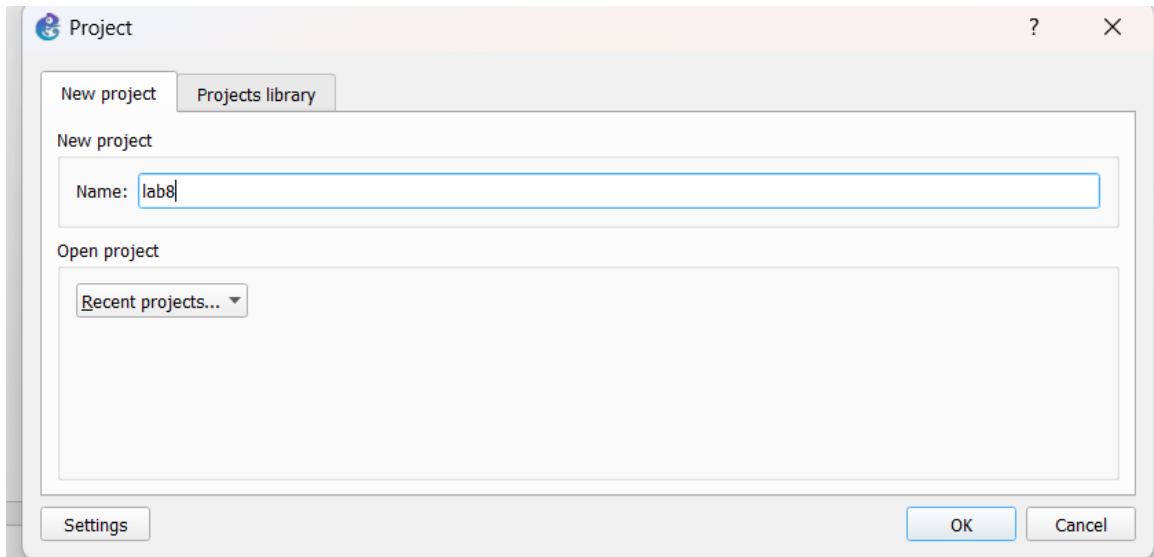
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1.2 Цель работы

Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования



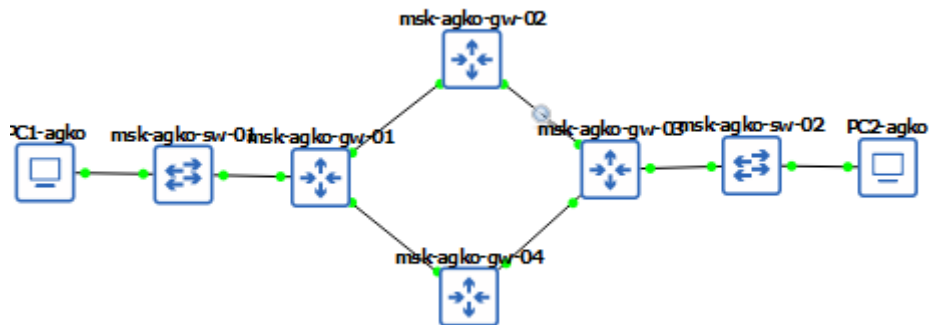


Рисунок 2: Топология сети в GNS3

```
PC1-agko> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
PC1-agko : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

PC1-agko> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
PC1-agko : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

PC1-agko> save
Saving startup configuration to startup.vpc
. done

PC1-agko> show ip

NAME          : PC1-agko[1]
IP/MASK       : 10.0.11.10/24
GATEWAY       : 10.0.11.1
DNS           :
MAC           : 00:50:79:66:68:00
LPORT        : 10016
RHOST:PORT    : 127.0.0.1:10017
```

```
Executing the startup file
```

```
PC2-agko> ip 10.0.11.10/24 10.0.11.1
```

```
Checking for duplicate address...
```

```
PC2-agko : 10.0.11.10 255.255.255.0 gateway 10.0.11.1
```

```
PC2-agko> save
```

```
Saving startup configuration to startup.vpc
```

```
. done
```

```
PC2-agko> show ip
```

```
NAME          : PC2-agko[1]  
IP/MASK       : 10.0.11.10/24  
GATEWAY       : 10.0.11.1  
DNS           :  
MAC           : 00:50:79:66:68:01  
LPORT        : 10066  
RHOST:PORT    : 127.0.0.1:10067
```

msk-agko-gw-01 - PuTTY

```
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-user-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
```

```
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-agko-gw-02# show running-config
Building configuration...

!
Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
exit
!
interface eth1
 ip address 10.0.2.1/24
exit
```

```
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
```



```
[OK]
msk-agko-gw-04# show running config
% Unknown command: show running config
msk-agko-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
```

 PC1-agko - PuTTY

```
PC1-agko> ip 2001:10:a/64
Invalid ipv6 address.

PC1-agko> ip 2001:10::a/64
PC1 : 2001:10::a/64

PC1-agko> save
Saving startup configuration to startup.vpc
. done

PC1-agko> show ipv6

NAME                : PC1-agko[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 2001:10::a/64
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:00
LPORT                : 10016
RHOST:PORT           : 127.0.0.1:10017
```

```
PC2-agko - PuTTY

Checking for duplicate address...
PC2-agko : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

PC2-agko> ip 2001:11::a/64
PC1 : 2001:11::a/64

PC2-agko> save
Saving startup configuration to startup.vpc
. done

PC2-agko> show ipv6

NAME                : PC2-agko[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10066
RHOST:PORT           : 127.0.0.1:10067
```

msk-agko-gw-01 - PuTTY

```
Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-01
service integrated-vtysh-config
!
interface eth0
    ipv6 address 2001:10::/64
    ipv6 address 2001:10::1/64
    no ipv6 nd suppress-ra
exit
!
interface eth1
    ipv6 address 2001:1::1/64
exit
!
interface eth2
    ipv6 address 2001:4::2/64
exit
```

```
msk-agko-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-02
service integrated-vtysh-config
!
interface eth0
  ipv6 address 2001:1::2/24
exit
!
interface eth1
  ipv6 address 2001:2::1/24
exit
!
interface eth-
```